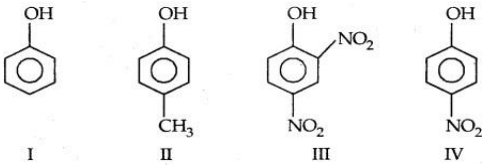


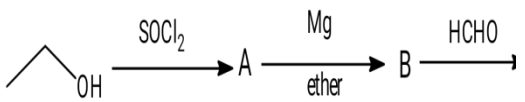
FREEDOM INTERNATIONAL SCHOOL

ALCOHOLS ,PHENOLS AND ETHERS

1. What happens when tertiary butyl alcohol is passed over heated copper at 300°C? (a) Secondary butyl alcohol is formed (b) 2-methylpropene is formed (c) 1-butene is formed (d) Butanol is formed	2. A compound X with the molecular formula C_3H_8O can be oxidized to another compound Y whose molecular formulae is $C_3H_6O_2$. The compound X may be (a) $CH_3CH_2OCH_3$ (b) CH_3CH_2CHO (c) $CH_3CH_2CH_2OH$ (d) $CH_3CHOHCH_3$
3. The decreasing order of boiling point of the following alcohols is (a) 3-methylbutan-2-ol > 2-methylbutan-2-ol > pentan-1-ol (b) Pentan-1-ol > 3-methylbutan-2-ol > 2-methylbutan-2-ol (c) 2-methylbutan-2-ol > 3-methylbutan-2-ol > pentan-1-ol (d) 2-methylbutan-2-ol > pentan-1-ol > 3-methylbutan-2-ol	4. Propanone on reaction with alkyl magnesium bromide followed by hydrolysis will produce (a) primary alcohol (b) secondary alcohol (c) tertiary alcohol (d) (d) carboxylic acid
5. Phenol when treated with excess of bromine water gives a white precipitate of (a) 2, 4, 6-tribromophenol (b) o-bromophenol (c) p-bromophenol (d) bromobenzene	6. The reaction between phenol and chloroform in the presence of aqueous NaOH is (a) nucleophilic substitution reaction (b) electrophilic addition reaction (c) electrophilic substitution reaction (d) nucleophilic addition reaction
7. The most suitable reagent for the conversion of $RCH_2OH \rightarrow RCHO$ is (a) $K_2Cr_2O_7$ (b) CrO_3 (c) $KMnO_4$ (d) PCC	8. Tertiary butyl alcohol gives tertiary butyl chloride on treatment with (a) $ConHCl$/anhydrous $ZnCl_2$ (b) KCN (c) $NaOCl$ (d) Cl_2
9. Conversion of phenol to salicylic acid and to salicylaldehyde are known as (respectively)	10. Benzoquinone is prepared by reaction of phenol with (a) $Na_2Cr_2O_7$, H_2SO_4

(a) Reimer-Tiemann reaction and Kolbe's reaction (b) Williamson's synthesis and Hydroboration-oxidation (c) Kolbe's reaction and Williamson's synthesis (d) Kolbe's reaction and Reimer-Tiemann reaction	(b) $\text{KMnO}_4, \text{H}_2\text{SO}_4$ (c) $\text{Na}_2\text{CrO}_4, \text{HCl}$ (d) $\text{K}_2\text{MnO}_4, \text{H}_2\text{SO}_4$
11. In the following reaction sequence Z is $\text{CH}_3 - \underset{\substack{ \\ \text{OH} \\ \text{(X)}}}{\text{CH}} - \text{CH}_3 \xrightarrow{[\text{O}]} \text{Y} \xrightarrow[\text{H}^+/\text{H}_2\text{O}]{\text{CH}_3\text{MgBr}} \text{Z}$ (a) butan-1-ol (b) butan-2-ol (c) 2-methylpropan-2-ol (d) 1, 1-dimethylethanol	12. Arrange the following alcohols in order of increasing reactivity towards sodium metal. (i) $(\text{CH}_3)_3\text{C}-\text{OH}$ (ii) $(\text{CH}_3)_2\text{CH}-\text{OH}$ (iii) $\text{CH}_3\text{CH}_2\text{OH}$ (a) (iii) < (ii) < (i) (b) (ii) > (i) < (iii) (c) (i) < (ii) < (iii) (d) (iii) < (i) < (ii)
13. An unknown alcohol is treated with "Lucas reagent" to determine whether the alcohol is primary, secondary or tertiary. Which alcohol reacts fastest and by what mechanism? (a) Tertiary alcohol by $\text{S}_{\text{N}}1$ (b) Secondary alcohol by $\text{S}_{\text{N}}1$ (c) Tertiary alcohol by $\text{S}_{\text{N}}2$ (d) Secondary alcohol by	14. An alcohol X when treated with hot cone. H_2SO_4 gave an alkene Y with formula C_4H_8 . This alkene on ozonolysis gives single product with molecular formula $\text{C}_2\text{H}_4\text{O}$. The alcohol is (a) butan-1-ol, (b) butan-2-ol (c) 2-methylpropan-1-ol (d) 2,2-dimethylbutanal-1-ol
15. The compound which gives the most stable carbonium ion on dehydration is (a) $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$ (b) $(\text{CH}_3)_3\text{COH}$ (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ (d) $\text{CH}_3\text{CH OH CH}_2\text{CH}_3$	16. Which of the following cannot be made by using Williamson Synthesis: (a) Methoxybenzene (b) Benzyl p-nitrophenyl ether (c) tert. butyl methyl ether (d) Ditert. butyl ether
17. The I.U.P.A.C. name of the ether	18. Strength of acidity is in order:

CH₂ = CH-CH₂O CH₃ is (a) Alkyl methyl ether (b) 1-Methoxy-2-propene (c) 3-Methoxy-1-propene (d) Vinyl dimethyl ether	 I II III IV (a) II > I > III > IV b) III > IV > I > II (b) I > IV > III > II d) IV > III > I > II
19. During dehydration of alcohols to alkenes by heating with cone. H₂SO₄ the initial step is (a) formation of an ester (b) protonation of alcohol molecule (c) formation of carbocation (d) elimination of water	20. $\text{HC} \equiv \text{CH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} \xrightarrow[\text{H}_2\text{O}]{\text{CH}_3\text{MgBr}} \xrightarrow{\text{PBr}_3}$ (a) CH₃CH Br CH₃ (b) CH₃ CH₂ CH₂ Br Br (c) CH₂ = CH-Br (d) Br CH = CH-CH₃
21. .When Phenol is distilled with zinc dust, it gives (a) Benzene (b) Toluene (c) Benzaldehyde (d) Benzoic acid	22. In the reaction of phenol with CHCl₃ and aqueous NaOH at 343 K, the electrophile attacking the ring is: (a) CHCl₃ (b) CHCl₂ (c) CCl₂ (d) COCl₂
23. Phenol reacts with Br₂ in CS₂ at low temperature to give (a) o-Bromophenol (b) o-and p-Bromophenols (c) p-Bromophenol (d) 2, 4, 6Tribromophenol	24. 1-Propanol and 2-propanol can be best distinguished by (a) Oxidation with KMnO₄ followed by reaction with Fehling solution? (b) Oxidation with acidic dichromate followed by reaction with Fehling solution. (c) Oxidation by heating with copper followed by reaction with Fehling solution. (d) Oxidation with cone. H₂SO₄ followed by reaction with Fehling solution.
25. Ethylene reacts with Baeyer's	26. By which of the following

reagent to give (a) ethane (b) ethyl alcohol (c) ethylene glycol (d) None of these	methods alcohol can be prepared in excellent yield? (a) From alkenes (b) By hydroboration-oxidation (c) From carbonyl compounds (d) From Grignard reagent
27. Identify D in the following reaction sequence  a)n-butylalcohol (b)Propanal (c)n-propylalcohol d) Butanal	28. Correct statements in case of n-butanol and t-butanol are: (i) both are having equal solubility in water (ii) t-butanol is more soluble in water than n-butanol (iii) boiling point of t-butanol is lower than n-butanol (iv) boiling point of n-butanol is lower than t-butanol (a) Only (iii) (b)Both (ii) and (iii) (c)Both (iii) and (iv) (d)(i), (iii) and (iv)
29. Which of the following forms the least stable carbocation? (a)(CH₃)₃COH (b)CH₂=CH-OH (c)CH₂=CHCH₂OH (d)C₆H₅CH₂OH	30. Though the molecular mass of ethanol and propane is same, the boiling point of ethanol is higher than propane, this is because (a)Ethanol is polar and propane is non polar (b)Ethanol involves in intermolecular hydrogen bonding but propane doesn't c)Both (a) and (b) (d)Only (b)